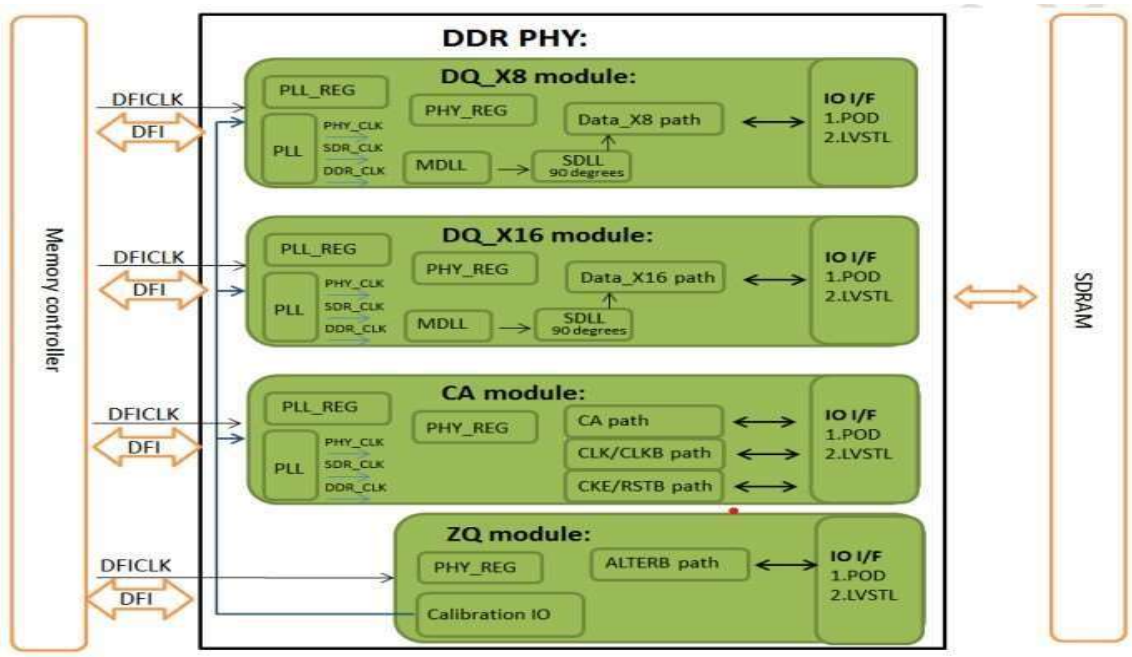
Overview

The DDR PHY IP supports DDR5/ DDR4/ LPDDR5, provides low latency, and enables up to 5400MT/s throughput. PHY functionality is verified inNC-Verilog simulation software using test bench written in Verilog HDL. The Combo PHY IP is also able to run on DDR4, DDR5,LPDDR5 modes separately.

Deliverables

- User Manual
- Behaviour model, and protected RTL codes
- Protected Post layout netlist
- Synopsys library (LIB)
- Frame view (LEF)
- Metal GDS (GDSII)
- Test patterns and Test Documentation

Features

- Supported DRAM type: DDR5/DDR4/LPDDR5
- Maximum controller clock frequency of 675MHz resulting in maximum DRAM data rate of 5400MT/s for DDR5
- Maximum controller clock frequency of 400MHz resulting in maximum DRAM data rate of 3200MT/s for DDR4
- Maximum controller clock frequency of 600MHz resulting in maximum DRAM data rate of 4800MT/s for LPDDR5
- Interface: POD11/POD12/LVSTL05
- Data path width scales in 8-bit increment
- Four modules for flexible configuration: CA/DQ_X16/DQ_X8/ZQ
- Programmable output impedance (DS)
- Programmable on-die termination (ODT)
- Core power:0.8V, Post-driver power (VDDQ):1.1V/1.2V/0.5V, Pre-driver power(VDDP): 1.1V/1.2V/1.05V
- Receiver power (VDDI): 1.1V/1.2V/1.05V for DDR5/DDR4/LPDDR5
- ESD: 2KV/HBM, 200V/MM, 500V/CDM
- Support ZQ calibration
- Support 4 ranks by each CA module
- Support write-leveling, CBT
- Support PHY internal VREFDQ auto decision
- Per-bit deskew in read and write datapath

Diagram